S4: generic degree four, specialized degree can drop

Lean checks the canonical S4 Sylvester resultant as an actual polynomial in each input position, over every field and in every characteristic. When the other three inputs remain independent formal variables, the degree in the chosen input is 4. After specializing those other inputs, the degree can drop. This preserves the qualified generic degree statement and refutes its unrestricted specialization strengthening. [pcc:unverified]

Let T occupy any one of the four input positions, with the other three coordinates denoted x, y, z. The checked statements are

$$\deg_T S_4 \le 4, \qquad [T^4]S_4 = S_3(x,y,z)^2,$$
$$\deg_T S_4 = 4 \iff S_3(x,y,z) \neq 0.$$

The definitions compute the actual nested-polynomial Sylvester determinant over the coefficient ring. Explicit map/evaluation lemmas connect every input position to the previously checked scalar S4. Polynomial equality is not inferred from finite-field pointwise agreement. [pcc:unverified]

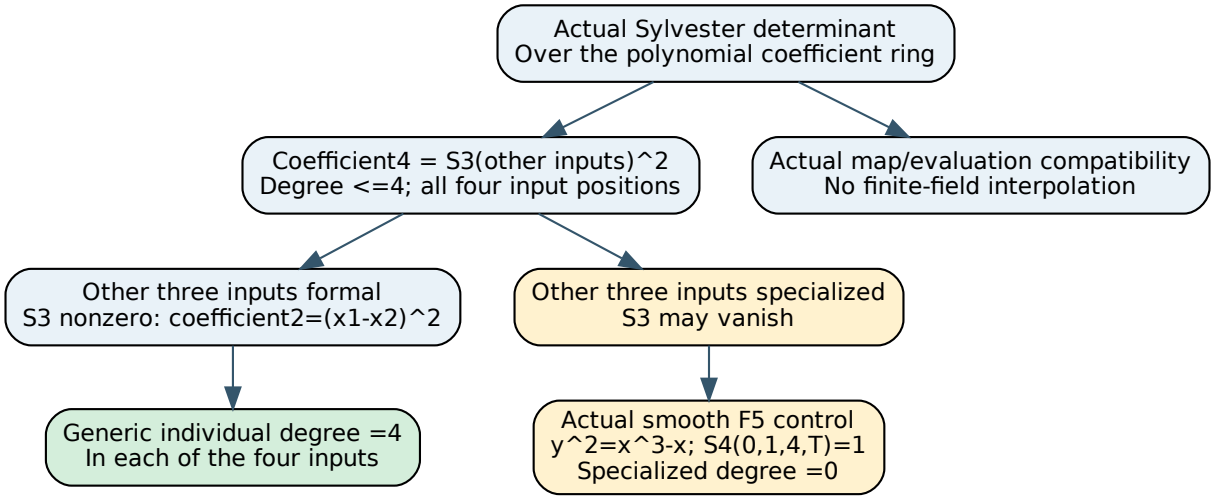


Figure 1: Canonical determinant, generic lower bound, and specialization control.

Each of the four symbolic presentations uses $K[x_1][x_2][x_3][T]$, with T in the selected source input position. The formal S3 is nonzero because its coefficient of the third indeterminate squared is $(x_1 - x_2)^2 \neq 0$. Pair swaps and the explicit symmetric S3 identity give the same leading coefficient criterion in every position. [pcc:unverified]

A smooth-curve specialization control

On the actual nonsingular curve $y^2 = x^3 - x$ over $\mathbf{F}_5$, Lean proves the following canonical identity. [pcc:unverified]

$$S_4(0,1,4,T) = 1, \qquad \deg_T S_4(0,1,4,T) = 0.$$

This is equality of canonical polynomials, not merely equality of their values on the five-element field. It disproves the assertion that fixing other inputs always preserves degree 4. It does not refute Semaev's generic individual-variable degree theorem. [pcc:unverified]

Situation	Checked conclusion
Three other coordinates formal and independent	Degree4 in each chosen position
Three coordinates specialized with S3 nonzero	Degree4
Three coordinates specialized with S3 zero	Degree below4
Smooth F5 model, last-variable triple(0,1,4)	Nonzero constant 1, degree 0

Source: ../../lean/S4Degree.lean. Main bindings include `positionPolynomial_curve_eval`, `positionPolynomial_coeff_four`, `positionPolynomial_curve_degree_iff`, `all_four_symbolic_degrees`, `actual_specialization_constant`, and `s4_degree_audit_certificate`. [pcc:unverified]

Whole census axiom gate: 259 declarations, 0 failing, 0 trusted-native exceptions; the count includes generated declarations, not 259 new results. Packet PCC-20261109-b93977 pins the explicit combined theorem. Its full local check passes17:28:09Z; standing remains unsigned. No custom axiom, proof hole or native_decide exception is accepted. [pcc:unverified]

Supplement IDC1h9d53585cac843775 checks only the scalar leading-term rewrite $\mathrm{Res}_{2,2}(aX^2+bX+c, (X-z)^2) = (az^2+bz+c)^2$: 32 derived and11 extra points over $2^{61}-1$,4 variables,degree bound 6. Local source commit 3d1dd96b preceded issue, and the proof-file digest matches. These one-host finite checks do not establish coefficient extraction, generic degrees, other characteristics or independent/signed standing; those universal mathematical conclusions rest on Lean. [pcc:unverified]

Primary source: Semaev2015, Section 2, pp2–3; PDF p3 visually checked for 2^{m-2} and algebraic-closure coordinates. Kosters–Yeo2015, Section 2/p3 supplies the general invariant formula/recursion. Exact read scopes and hashes: ../../s4-degree-sources.json. Original Semaev 2004 remains unread after 403/404 responses. These are not full-paper or all-fourteen audits. [pcc:unverified]

Scope is $m = 4$. General all-m recursion/degree, irreducibility and the remaining requested obligations stay open. No solving-degree, regularity or algorithm-complexity conclusion follows from this polynomial degree. No new native curve population, benchmark, frozen-data change or paid compute was run. All 11 standalone attempt logs and the final gate are retained. [pcc:unverified]